NUTRITION

Community Medicine (PSM) | 3rd Year

Hi Sammy!
Nutrition = guaranteed
20+ marks in PSM!
Tables are KEY here!

Nutrition | PSM | 3rd Year | MBBS

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Know your macro vs micronutrients and BMI classification cold!

Classification of Nutrients

Category	Nutrients	Key Points
Macronutrients	Carbohydrates	4 kcal/g — main energy source (60-70% diet)
	Proteins	4 kcal/g — body building; 1 g/kg/day (adult)
	Fats	9 kcal/g — highest energy; EFA = linoleic, linolenic
Micronutrients	Vitamins	Fat-soluble (ADEK) + Water-soluble (B, C)
	Minerals	Ca, Fe, I, Zn, Se — deficiencies common in India
Other	Water	2-3 L/day; 1 mL/kcal energy expenditure
	Dietary fibre	25-40 g/day; prevents constipation, CVD, diabetes

Protein Classification by Quality

Type	Definition	Examples
Complete (1st class)	All essential amino acids	Egg, milk, meat, fish, soy
Partially incomplete	Low in 1-2 EAAs	Cereals (low lysine), pulses (low met)
Incomplete (3rd class)	Lack several EAAs	Gelatin

★ **MUTUAL SUPPLEMENTATION: Cereal + Pulse = complete protein!**

Dal-chawal / roti-dal = Indian diet wisdom! (Lysine + Methionine complement)

BMI Classification (WHO / Asian)

Category	WHO BMI	Asian BMI	Risk
Underweight	< 18.5	< 18.5	Malnutrition risk
Normal	18.5 - 24.9	18.5 - 22.9	Low risk
Overweight	25.0 - 29.9	23.0 - 24.9	Increased risk
Obese Class I	30.0 - 34.9	25.0 - 29.9	Moderate risk
Obese Class II	35.0 - 39.9	> 30.0	Severe risk
Obese Class III	> 40.0	—	Very severe risk

★ **BMI = Weight (kg) / Height (m)^2**

Asian cut-offs are LOWER because Asians have higher body fat at same BMI!

This comparison table is THE most important in all of nutrition!

PEM = Protein Energy Malnutrition

Spectrum: Underweight -> Marasmus -> Kwashiorkor

Marasmic-Kwashiorkor = combined form (worst prognosis).

Kwashiorkor vs Marasmus — THE Table

Feature	Kwashiorkor	Marasmus
Deficiency	Protein (mainly)	Calories + Protein (total energy)
Age group	1-3 years (after weaning)	< 1 year (infancy)
Oedema	PRESENT (hallmark)	Absent
Muscle wasting	Present but masked by oedema	Severe — skin and bones
Weight	60-80% of expected	< 60% of expected
Skin changes	Flaky paint / crazy pavement	Loose, wrinkled, baggy pants
Hair changes	Flag sign, sparse, depigmented	Thin, sparse, easily pluckable
Appetite	Poor (anorexia)	Good (ravenously hungry)
Face	Moon face (puffy)	Old man / monkey face
Liver	Fatty liver (hepatomegaly)	No fatty change
Serum albumin	Very LOW (< 2.5 g/dL)	Low-normal
Mental changes	Apathetic, irritable, miserable	Alert but irritable

IAP Classification of PEM (Indian Pediatrics)

Grade	Weight for Age (% of median)	Clinical Significance
Normal	> 80%	No malnutrition
Grade I (Mild)	71-80%	Underweight — monitor closely
Grade II (Moderate)	61-70%	Nutritional supplementation needed
Grade III (Severe)	51-60%	Severe PEM — medical management
Grade IV (Very severe)	< 50%	Marasmus — hospital admission

★ **K = Kwashiorkor = Keep Oedema (K for oedema Key feature)**

M = Marasmus = Muscle wasting (M = Muscles Melted). Flag sign = only Kwashiorkor!

Vitamin Deficiency Diseases

Vitamin deficiencies = MCQ goldmine! Learn the associations cold!

Fat-Soluble Vitamins (A, D, E, K)

Vitamin	Source	Deficiency Disease	Key Feature
A	Liver, carrot, mango	Xerophthalmia (night blindness)	Bitot's spots, keratomalacia
D	Sunlight, fish, egg	Rickets (child), Osteomalacia	Craniotabes, rachitic rosary
E	Veg oils, nuts, wheat	Hemolytic anaemia (neonates)	Antioxidant; infertility
K	Green leafy, gut flora	Hemorrhagic disease (newborn)	Needed for II, VII, IX, X

Water-Soluble Vitamins (B complex, C)

Vitamin	Source	Deficiency Disease	Key Feature
B1	Whole grains, pulses	Beriberi (wet/dry)	Wet=heart; Dry=neuro; Wernicke
B2	Milk, eggs, liver	Ariboflavinosis	Angular stomatitis, glossitis
B3	Meat, pulses, groundnut	Pellagra (3 Ds)	Dermatitis, Diarrhea, Dementia
B6	Meat, banana, potato	Peripheral neuropathy	INH-induced deficiency
B9	Green leafy, liver	Megaloblastic anaemia	Neural tube defects (NTD)
B12	Animal products only	Megaloblastic + neuro	Subacute combined degen.
C	Citrus, amla, guava	Scurvy	Bleeding gums, perifollicular hem.

Vitamin A Prophylaxis — National Schedule

Dose	Age	Details
1 lakh IU	9 months (with measles)	First dose with measles vaccine
2 lakh IU	16-18 months	Second dose
2 lakh IU	Every 6 months	Up to 5 years of age (total 9 doses)

Food Fortification in India

Food Vehicle	Fortified With	Programme / Detail
Salt	Iodine	National Iodine Deficiency Disorders CP
Salt (double)	Iodine + Iron	Double-fortified salt (DFS) programme
Wheat flour (atta)	Iron + Folic acid	FSSAI mandated fortification
Rice	Iron + B12 + Folic acid	FCI fortified rice for PDS
Edible oil	Vitamin A + D	FSSAI standards
Milk	Vitamin A + D	Amul and other dairies

ICDS, POSHAN, MDM =
Indian context questions
come in EVERY exam!

ICDS — Integrated Child Development Services

Launched:	1975 (2nd October) — world's largest programme for children
Target:	Children < 6 yrs, pregnant/lactating women, adolescent girls
Functionary:	Anganwadi Worker (AWW) at Anganwadi Centre (AWC)
Services:	6 services — supplementary nutrition, immunisation, health checkup, referral, nutrition & health education, preschool education
Supplement:	Calorie: 500 kcal + Protein: 12-15 g/day (children 1-6 yrs)
Special:	Severely malnourished: 800 kcal + 20-25 g protein/day

PM-POSHAN (Mid-Day Meal Scheme)

Launched:	1995 (National Programme); renamed PM-POSHAN in 2021
Target:	Children in classes 1-8 in govt/aided schools
Calorie:	Primary: 450 kcal, 12g protein | Upper primary: 700 kcal, 20g protein
Objective:	Improve nutrition, increase school enrollment & attendance

POSHAN Abhiyaan (National Nutrition Mission)

Launched:	2018 (March 8, International Women's Day)
Target:	Reduce stunting from 38.4% to 25% by 2022 (NFHS-4 baseline)
Focus:	Stunting, under-nutrition, anaemia (women & children), low birth weight
Strategy:	Technology (ICDS-CAS app), convergence, behaviour change

Other Important Nutrition Programmes

Programme	Key Detail
National Iodine Deficiency CP	Iodised salt — universal salt iodisation since 1992
National Iron+ Initiative	IFA supplementation across all age groups
Wheat Flour Fortification	Iron + Folic acid added to atta (FSSAI)
Saksham Anganwadi (Mission)	Upgraded ICDS with technology & infrastructure

Nutrition is your BEST friend in PSM exams! Go get those marks!

Most Repeated LAQs (10–12 marks)

1. Differentiate Kwashiorkor vs Marasmus (table + clinical features)
2. Describe ICDS — objectives, services, functionaries, beneficiaries
3. Classify vitamins. Describe Vitamin A deficiency and national programme
4. Write about Protein Energy Malnutrition — classification, clinical, management
5. Describe national nutrition programmes in India (ICDS, MDM, POSHAN)

Most Repeated SAQs (5–6 marks)

1. BMI classification (WHO and Asian cut-offs)
2. Pellagra — 3 Ds, cause, treatment
3. Vitamin A prophylaxis schedule
4. Food fortification in India
5. Mid-Day Meal Scheme — objectives and calorie norms
6. IAP classification of PEM

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Oedema + flaky paint dermatosis	Kwashiorkor
Monkey face / old man face	Marasmus
Flag sign of hair	Kwashiorkor
Bitot's spots	Vitamin A deficiency
Rachitic rosary	Vitamin D deficiency (Rickets)
3 Ds: Dermatitis, Diarrhea, Dementia	Pellagra (Niacin / B3 deficiency)
Scurvy	Vitamin C deficiency
Bleeding disorder in newborn	Vitamin K deficiency
ICDS functionary	Anganwadi Worker (AWW)
World's largest child nutrition programme	ICDS (India)
Mutual supplementation example	Dal + Chawal (cereal + pulse)
Reference protein (highest BV)	Egg (Biological Value = 100)

♥ **Good nutrition builds strong nations, Sammy!**
You're going to help so many children grow healthy!